

A Finite Template Reduction for Minimal Balanced Collections

Partial progress on the enumeration problem

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Abstract

Let $N = [n]$. A collection $\mathcal{B} \subseteq 2^N \setminus \{\emptyset\}$ is balanced if there are strictly positive weights λ_S such that every player is covered with total weight one. It is minimal if no proper subcollection is balanced. The enumeration of minimal balanced collections is a recurring obstruction in the geometry of balanced games. This note gives a rigorous finite reduction of that enumeration to a family of Boolean templates. More precisely, ordered minimal balanced collections with k coalitions are in bijection with admissible subsets $C \subseteq \{0, 1\}^k \setminus \{0\}$ of rank k , together with a surjection from the player set onto C . Consequently

$$|\mathcal{B}^*(n)| = \sum_{k=2}^n \frac{1}{k!} \sum_{C \in \mathfrak{T}_k} \text{Sur}(n, |C|) = \sum_{k=2}^n \frac{1}{k!} \sum_q t_{kq} q! \left\{ \begin{matrix} n \\ q \end{matrix} \right\}.$$

Here $\mathfrak{T}_k$ is a finite, explicitly testable set of templates and $t_{kq} = |\{C \in \mathfrak{T}_k : |C| = q\}|$. The result should be regarded as partial progress rather than a solution of the general enumeration problem, because the constants t_{kq} are not evaluated in closed form.

1 Status of the contribution

This note does *not* claim a full solution of the open enumerative problem for minimal balanced collections. It proves a canonical reduction: every minimal balanced collection is obtained by “blowing up” a finite Boolean template by a surjection from the labelled player set. This gives an exact formula once the template counts t_{kq} are known, and it gives a finite exact test for whether a proposed template is admissible. The unresolved part is the enumeration of those admissible templates themselves. Thus the contribution is best described as a correct reformulation and finite reduction, not a closed enumeration, not an asymptotic formula, and not a complete solution.

The motivation is the role of minimal balanced collections in cooperative game theory. Bondareva–Shapley balancedness characterizes games with nonempty core [1, 5]; minimal balanced collections appear in the facet structure of the cone of balanced games studied by Garcia-Segador, Grabisch, and Miranda [2]. The difficulty of generating and counting minimal balanced collections is discussed explicitly in [4, 3].

2 Definitions and notation

Fix $N = [n] = \{1, \dots, n\}$. A collection of coalitions is a finite set

$$\mathcal{B} \subseteq 2^N \setminus \{\emptyset\}.$$

If $\mathcal{B} = \{S_1, \dots, S_k\}$ is temporarily ordered, its incidence matrix is the matrix

$$A_{\mathcal{B}} \in \{0, 1\}^{n \times k}, \quad (A_{\mathcal{B}})_{ij} = \begin{cases} 1, & i \in S_j, \\ 0, & i \notin S_j. \end{cases}$$

The collection $\mathcal{B}$ is *balanced* if there exists a vector $\lambda \in \mathbb{R}_{\geq 0}^k$ such that

$$A_{\mathcal{B}}\lambda = \mathbf{1}_n. \quad (1)$$

Equivalently,

$$\sum_{S \in \mathcal{B}: i \in S} \lambda_S = 1 \quad \text{for every } i \in N.$$

A balanced collection is *minimal balanced* if no proper subcollection is balanced.

Let $\mathcal{B}^*(n)$ denote the set of nontrivial minimal balanced collections on $[n]$, where the trivial one-element collection $\{[n]\}$ is excluded. This convention agrees with the usual role of $\mathcal{B}^*(n)$ in the facet description of balanced games. Notice that if $[n] \in \mathcal{B}$ and $|\mathcal{B}| > 1$, then $\{[n]\}$ is already balanced, so $\mathcal{B}$ cannot be minimal. Thus the only minimal balanced collection containing the grand coalition is the excluded trivial one.

We use

$$\text{Sur}(n, q) = q! \begin{Bmatrix} n \\ q \end{Bmatrix} = \sum_{r=0}^q (-1)^r \binom{q}{r} (q-r)^n$$

for the number of surjections from an n -element set onto a q -element set, with the convention $\text{Sur}(n, q) = 0$ for $q > n$.

3 The rank criterion for minimality

The first observation is that, for balanced collections, minimality is exactly full column rank of the incidence matrix.

Lemma 3.1 (Rank criterion). *Let $\mathcal{B} = \{S_1, \dots, S_k\}$ be balanced, and let $A = A_{\mathcal{B}}$. Then $\mathcal{B}$ is minimal balanced if and only if*

$$\text{rank } A = k.$$

Consequently every minimal balanced collection on $[n]$ contains at most n coalitions and has a unique balancing vector.

Proof. Assume first that $\text{rank } A = k$. Let $\lambda \in \mathbb{R}_{\geq 0}^k$ satisfy $A\lambda = \mathbf{1}_n$. If a proper subcollection were balanced, then by inserting zero coordinates in the positions of the omitted coalitions we would obtain a vector $\mu \in \mathbb{R}_{\geq 0}^k$, with at least one zero coordinate, satisfying $A\mu = \mathbf{1}_n$. Hence

$$A(\lambda - \mu) = 0.$$

Since A has full column rank, its kernel is zero, so $\lambda = \mu$. This contradicts $\lambda > 0$ and the fact that μ has a zero coordinate. Thus $\mathcal{B}$ is minimal.

Conversely, suppose that $\text{rank } A < k$. Choose a nonzero vector $z \in \ker A$. The vector z cannot be coordinatewise nonnegative: if $z \geq 0$ and $z_j > 0$ for some j , then column j of A is nonzero, so Az has a positive entry. Similarly z cannot be coordinatewise nonpositive. Hence z has at least one positive and at least one negative coordinate.

Let $\lambda \in \mathbb{R}_{>0}^k$ satisfy $A\lambda = \mathbf{1}_n$. Consider the closed interval

$$I = \{t \in \mathbb{R} : \lambda + tz \geq 0\}.$$

Because z has both signs, I is a compact interval with 0 in its interior. At either endpoint t_0 of I , the vector

$$\mu = \lambda + t_0 z$$

is nonnegative and has at least one zero coordinate. It is not the zero vector, since z is not a scalar multiple of the positive vector λ with the opposite sign. Also

$$A\mu = A\lambda + t_0 Az = \mathbf{1}_n.$$

Restricting μ to its positive coordinates therefore gives strictly positive balancing weights for a nonempty proper subcollection of $\mathcal{B}$. Hence $\mathcal{B}$ is not minimal.

The final assertions follow immediately: full column rank gives $k \leq n$, and it also gives uniqueness of the solution of (1). $\square$

4 Boolean templates

For $k \geq 1$, put

$$\mathcal{Q}_k = \{0, 1\}^k \setminus \{0\}.$$

A vector $c = (c_1, \dots, c_k) \in \mathcal{Q}_k$ records a player type: $c_j = 1$ means that a player of type c belongs to the j th coalition. If $C \subseteq \mathcal{Q}_k$, then $\text{rank } C$ means the row rank of the matrix whose rows are the elements of C .

Definition 4.1 (Admissible template). *A subset $C \subseteq \mathcal{Q}_k$ is an admissible k -template if*

(i) $\text{rank } C = k$;

(ii) *there exists a vector $\lambda \in \mathbb{R}_{>0}^k$ such that*

$$c \cdot \lambda = 1 \quad \text{for every } c \in C. \tag{2}$$

Let $\mathfrak{T}_k$ be the finite set of admissible k -templates.

The rank condition in Definition 4.1 is essential. It is the template-level version of minimality, whereas (2) is the template-level version of balancedness.

5 The template parametrization

The next theorem is the main result of the note. It says that the only freedom in passing from templates to labelled players is the choice of a surjection from players onto row types.

Theorem 5.1 (Template parametrization). *Fix $n \geq 1$ and $1 \leq k \leq n$. Ordered minimal balanced collections*

$$(S_1, \dots, S_k)$$

with k coalitions on $[n]$ are in bijection with pairs (C, ϕ) consisting of an admissible template $C \in \mathfrak{T}_k$ and a surjection

$$\phi : [n] \twoheadrightarrow C.$$

The pair (C, ϕ) corresponds to the ordered collection defined by

$$S_j = \{i \in [n] : \phi(i)_j = 1\}, \quad j = 1, \dots, k.$$

For $k = 1$ this construction gives exactly the excluded trivial collection $\{[n]\}$.

Proof. Start with an ordered minimal balanced collection $(S_1, \dots, S_k)$ and let $A = A_{\mathcal{B}}$. Let $\lambda \in \mathbb{R}_{>0}^k$ be its balancing vector. For each player $i \in [n]$, define the row type

$$\rho(i) = (\mathbf{1}_{\{i \in S_1\}}, \dots, \mathbf{1}_{\{i \in S_k\}}) \in \{0, 1\}^k.$$

The vector $\rho(i)$ is nonzero, because

$$\rho(i) \cdot \lambda = 1.$$

Set

$$C = \rho([n]) \subseteq \mathcal{Q}_k.$$

Then $\rho : [n] \rightarrow C$ is surjective by definition. Every row type $c \in C$ satisfies $c \cdot \lambda = 1$. Moreover, the row span of A is exactly the span of the distinct row types in C ; hence

$$\text{rank } C = \text{rank } A = k$$

by Lemma 3.1. Thus $C \in \mathfrak{T}_k$, and we have constructed a pair (C, ρ) .

Conversely, let $C \in \mathfrak{T}_k$ and let $\phi : [n] \rightarrow C$ be a surjection. Choose $\lambda \in \mathbb{R}_{>0}^k$ satisfying (2), and form the $n \times k$ matrix A whose i th row is $\phi(i)$. Then every row of A has dot product one with λ , so

$$A\lambda = \mathbf{1}_n.$$

Since ϕ is surjective, every row of C occurs among the rows of A . Therefore

$$\text{rank } A = \text{rank } C = k.$$

By Lemma 3.1, the collection defined by the columns of A is minimal balanced. Full column rank also implies that no column is zero and no two columns are equal, so these columns define k distinct nonempty coalitions.

The two constructions are inverse to one another: beginning with an ordered collection, taking its distinct row types, and then reconstructing the columns recovers the original ordered collection; beginning with (C, ϕ) and taking row types recovers C and ϕ . The case $k = 1$ has only the template $C = \{(1)\}$ and gives the one-element collection $\{[n]\}$. $\square$

Remark 5.2 (Unordered collections). *The theorem is stated for ordered collections only to avoid quotient notation. A nontrivial minimal balanced collection has distinct coalitions, so it has exactly $k!$ orderings. Therefore unordered collections are counted by dividing the ordered count by $k!$.*

6 Counting formulas

Corollary 6.1 (Exact formula in terms of templates). *For every $n \geq 1$,*

$$|\mathcal{B}^*(n)| = \sum_{k=2}^n \frac{1}{k!} \sum_{C \in \mathfrak{T}_k} \text{Sur}(n, |C|). \quad (3)$$

Equivalently, if

$$t_{kq} = |\{C \in \mathfrak{T}_k : |C| = q\}|,$$

then

$$|\mathcal{B}^*(n)| = \sum_{k=2}^n \frac{1}{k!} \sum_{q=k}^{\min(n, 2^k-1)} t_{kq} q! \left\{ \begin{matrix} n \\ q \end{matrix} \right\}. \quad (4)$$

Proof. Fix k . By Theorem 5.1, for each admissible template $C \in \mathfrak{T}_k$, ordered minimal balanced collections with row-type set C are counted by surjections $[n] \twoheadrightarrow C$, namely by $\text{Sur}(n, |C|)$. Summing over $C \in \mathfrak{T}_k$ gives the number of ordered collections with k coalitions. By Remark 5.2, each unordered collection is counted exactly $k!$ times. Finally, $k = 1$ is the excluded trivial collection and Lemma 3.1 gives $k \leq n$ for every nontrivial minimal balanced collection. $\square$

The exponential generating function form is often the cleanest way to summarize the same reduction:

$$\sum_{n \geq 0} |\mathcal{B}^*(n)| \frac{x^n}{n!} = \sum_{k \geq 2} \frac{1}{k!} \sum_{C \in \mathfrak{T}_k} (e^x - 1)^{|C|}, \quad (5)$$

where the identity is interpreted coefficientwise. For each coefficient of x^n , only the terms with $k \leq |C| \leq n$ contribute.

7 A determinant test for templates

The formulas above are finite because admissibility of a template is decidable by exact determinant tests over 0-1 matrices.

Proposition 7.1 (Determinantal membership test). *Let $C = \{c_1, \dots, c_q\} \subseteq \mathcal{Q}_k$. Then $C \in \mathfrak{T}_k$ if and only if there exists a subset $I = \{i_1, \dots, i_k\} \subseteq [q]$ such that the $k \times k$ matrix*

$$M = \begin{pmatrix} c_{i_1} \\ \vdots \\ c_{i_k} \end{pmatrix}$$

is nonsingular, the vector

$$\lambda = M^{-1} \mathbf{1}_k$$

has strictly positive coordinates, and

$$c_\ell M^{-1} \mathbf{1}_k = 1 \quad (\ell = 1, \dots, q). \quad (6)$$

Equivalently, $C \in \mathfrak{T}_k$ if and only if such an M exists and the following determinant tests hold:

$$\frac{\det M^{(j)}}{\det M} > 0 \quad (j = 1, \dots, k), \quad (7)$$

where $M^{(j)}$ is obtained from M by replacing its j th column by $\mathbf{1}_k$, and

$$\det \begin{pmatrix} M & \mathbf{1}_k \\ c_\ell & 1 \end{pmatrix} = 0 \quad (\ell = 1, \dots, q). \quad (8)$$

Proof. If $C \in \mathfrak{T}_k$, then $\text{rank } C = k$, so some k rows form a nonsingular matrix M . The equations $M\lambda = \mathbf{1}_k$ force

$$\lambda = M^{-1}\mathbf{1}_k.$$

Thus admissibility is exactly the conjunction of positivity of this vector and the remaining row equations (6).

Conversely, if such an M exists and the vector $\lambda = M^{-1}\mathbf{1}_k$ is positive and satisfies (6), then all rows of C satisfy $c \cdot \lambda = 1$, and the nonsingularity of M gives $\text{rank } C = k$. Hence $C \in \mathfrak{T}_k$.

The determinant formulation follows from Cramer's rule,

$$\lambda_j = \frac{\det M^{(j)}}{\det M},$$

and from the Schur-complement identity

$$\det \begin{pmatrix} M & \mathbf{1}_k \\ c_\ell & 1 \end{pmatrix} = \det(M)(1 - c_\ell M^{-1}\mathbf{1}_k).$$

Since M is nonsingular, the determinant in (8) vanishes exactly when $c_\ell M^{-1}\mathbf{1}_k = 1$. □

Remark 7.2 (What the determinant test does and does not do). *Proposition 7.1 gives an exact finite test for a proposed template C . It therefore yields a finite algorithm for computing $|\mathcal{B}^*(n)|$ for any fixed n . It does not, by itself, give a closed formula or useful asymptotic estimate for the template numbers t_{kq} . This is the main reason that Corollary 6.1 should be viewed as a reduction rather than a complete enumeration.*

8 Reconstruction algorithm for a fixed player set

For fixed n , the preceding results give a conceptually simple exact generation algorithm.

1. For each $k = 2, \dots, n$, enumerate subsets $C \subseteq \mathcal{Q}_k$ with $k \leq |C| \leq n$.
2. Retain exactly those C that pass the determinant test in Proposition 7.1.
3. For each retained C , enumerate all surjections $\phi : [n] \twoheadrightarrow C$.
4. Output the collection

$$S_j = \{i \in [n] : \phi(i)_j = 1\}, \quad j = 1, \dots, k,$$

and identify outputs that differ only by a permutation of the k coalitions.

Correctness is exactly Theorem 5.1; the number of outputs is exactly Corollary 6.1. This algorithm is not presented as computationally efficient. Its value is that it separates the labelled-player part of the problem, which is just a surjection count, from the genuinely structural part, which is the enumeration of admissible Boolean templates.

9 Small checks

The smallest template sets are

$$\mathfrak{T}_1 = \{\{(1)\}\}, \quad \mathfrak{T}_2 = \{\{(1,0), (0,1)\}\},$$

and

$$\mathfrak{T}_3 = \{\{(1,0,0), (0,1,0), (0,0,1)\}, \\ \{(1,1,0), (1,0,1), (0,1,1)\}\}.$$

For $k = 4$, exact determinant enumeration gives

$$t_{4,4} = 22, \quad t_{4,5} = 6, \quad t_{4,6} = 1,$$

with all other $t_{4,q}$ equal to zero. For the value at $n = 5$, one also needs

$$t_{5,5} = 976.$$

Then (4) gives

$$|\mathcal{B}^*(1)| = 0, \quad |\mathcal{B}^*(2)| = 1, \quad |\mathcal{B}^*(3)| = 5, \quad |\mathcal{B}^*(4)| = 41, \quad |\mathcal{B}^*(5)| = 1291.$$

For example, the $n = 5$ count decomposes by number of coalitions as

$$15 + 50 + 250 + 976 = 1291,$$

corresponding to $k = 2, 3, 4, 5$ respectively.

10 Conclusion

The main theorem gives a canonical row-type decomposition of minimal balanced collections. In this decomposition, the labelled players enter only through a surjection onto the set of row types. This proves the exact template formula (3) and the equivalent formula (4).

The result is nevertheless only partial progress on the broad enumeration problem. The difficult problem is transformed into the enumeration of admissible Boolean templates $\mathfrak{T}_k$, or equivalently the constants t_{kq} . Without a closed description, recurrence, asymptotic analysis, or efficient enumeration of those constants, the formula above should not be advertised as a full solution of the authors' problem. Its contribution is a rigorous finite reduction and a clean separation between template structure and labelled-player blow-ups.

References

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